

Chapter 12, Question 6

Question: Given the first M principal component score vectors z_{im} ($m = 1, \dots, M$), explain, using equation (12.6), that each of the first M principal component loading vectors ϕ_{jm} can be obtained by performing p separate least squares linear regressions, in which the scores are the predictors and one feature of the data matrix is the response.

Answer

Equation (12.6) is:

$$\text{minimize over } A, B: \sum_j \sum_i (x_{ij} - \sum_m a_{im} b_{jm})^2$$

where A is $n \times M$ (the scores) and B is $p \times M$ (the loadings). Now suppose the score vectors z_{im} are already known and fixed — only the loadings b_{jm} are left to find.

Split the double sum by feature j . Since each column j of X contributes an independent term to the total sum of squares, the overall minimization decomposes into p separate, independent minimization problems — one per feature:

$$\text{For each fixed } j: \text{minimize over } b_{j1}, \dots, b_{jM}: \sum_i (x_{ij} - \sum_m z_{im} b_{jm})^2$$

Look closely at this inner problem. For a fixed feature j , x_{ij} ($i = 1, \dots, n$) is just a single response vector — the values of that one feature across all n observations. And $\sum_m z_{im} b_{jm}$ is a linear combination of the M known score vectors $z_{i1}, \dots, z_{iM}$, with $b_{j1}, \dots, b_{jM}$ as the unknown coefficients.

That is exactly the least squares regression problem: regress the response x_j (feature j across all observations) onto the M predictors $z_1, \dots, z_M$, with no intercept (since both the scores and the original data are centered to have mean zero). The coefficients that solve this regression are precisely $\phi_{j1}, \dots, \phi_{jM}$ — the loadings for feature j .

Since this holds independently for every $j = 1, \dots, p$, the full set of loadings can be recovered by running p separate least squares regressions, each one:

$$x_j \sim z_1 + z_2 + \dots + z_M \text{ (no intercept)}$$

and reading off the p sets of regression coefficients as the M loading vectors.

Why this makes sense

This is really just the flip side of how PCA is usually described. Normally we think of loadings as fixed and scores as unknowns to solve for (finding the direction of maximum variance). Equation (12.6) is symmetric in A and B , so if we instead fix the scores and ask "what loadings best reconstruct the data from these scores," the answer is exactly an ordinary least squares fit — because (12.6) is nothing more than a sum of squared residuals, and minimizing a sum of squared residuals with known predictors and an unknown coefficient vector is the definition of least squares.